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Functional Insights Into the Effect of Feralisation on the Gut Microbiota of Cats Worldwide

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ABSTRACT

Successfully adapting to a feral lifestyle with different access to food, shelter and other resources requires rapid physiological and behavioural changes, which could potentially be facilitated by gut microbiota plasticity. To investigate whether alterations in gut microbiota support this transition to a feral lifestyle, we analysed the gut microbiomes of domestic and feral cats from six geographically diverse locations using genome-resolved metagenomics. By reconstructing 229 non-redundant metagenome-assembled genomes from 92 cats, we identified a typical carnivore microbiome structure, with notable diversity and taxonomic differences across regions. While overall diversity metrics did not differ significantly between domestic and feral cats, hierarchical modelling of species communities, accounting for geographic and sex covariates, revealed significantly larger microbial functional capacities among feral cats. The increased capacity for amino acid and lipid degradation corresponds to feral cats' dietary reliance on crude protein and fat. A second modelling analysis, using behavioural phenotype as the main predictor, unveiled a positive association between microbial production of short-chain fatty acids, neurotransmitters and vitamins and cat aggressiveness, suggesting that gut microbes might contribute to heightened aggression and elusiveness observed in feral cats. Functional microbiome shifts may therefore play a significant role in the development of physiological and behavioural traits advantageous for a feral lifestyle, a hypothesis that warrants validation through microbiota manipulation experiments.

1 | Introduction

Feralisation is the process by which a once-domesticated organism detaches from the anthropic environment (Henriksen et al. 2018). While possessing some distinct characteristics, feralisation can generally be perceived as the counterpoint to domestication (E. O. Price 1984). Many species initially domesticated by humans have subsequently given rise to feral populations (Gering et al. 2019), often leading to adverse impacts on

both human settlements and biodiversity (Bonacic et al. 2019; Medina et al. 2011; Palmas et al. 2017). Despite the profound implications of this phenomenon, feralisation remains a relatively understudied process compared to its counterpart, domestication.

One of the animal species that is commonly found in feral form is the house cat (*Felis silvestris catus*). The domestication of cats is thought to have taken place in the Near East at the

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onset of the Neolithic period (Driscoll et al. 2007). This likely occurred concurrently with the rise of agriculture, as cats encountered prey hotspots around grain stores, while becoming valuable assets for humans in controlling rodent populations (Clutton-Brock 1990; Driscoll et al. 2009; Gross 2020). Since then, cats have spread alongside humans across most of the planet (Baca et al. 2018; Ottoni and Van Neer 2020). In every region they have colonised, cats have demonstrated an exceptional ability to thrive outside of domestic settings, forming countless feral cat populations worldwide (Bradshaw et al. 1999). Therefore, while domestication happened in a specific geographic area and time point, feralisation has happened and is continuously happening all over the world. This is partly due to the fact that modern non-pedigree domestic cats still closely resemble their wild ancestors both genetically and morphologically, retaining a significant portion of their wild behavioural repertoire, including proficient hunting skills (Bradshaw 2006; Cecchetti et al. 2021). Additionally, feral cats typically display behaviours more akin to their wild relatives, such as an aversion to humans (Levy et al. 2003). The feralisation of cats poses a significant threat to the environment, as feral cats have been implicated as the primary cause of at least 14% of global bird, mammal, and reptile extinctions (Dueñas et al. 2021; Medina et al. 2011).

Feralisation success is often linked to phenotypic plasticity (Gering et al. 2019), as feral animals must adapt to significantly more variable and unpredictable environments compared to their domestic counterparts. In this regard, there is growing interest in the potential role that the gut microbiota may play in assisting such transitions. In general, exposure to new environments can lead to the acquisition of novel intestinal microorganisms, and these can interact with the host in diverse ways (Candela et al. 2012). While traditional microbiological and veterinary research has largely concentrated on pathogens, there is an increasing recognition of the beneficial effects of gut microorganisms (Lee and Hase 2014; Ma et al. 2023). Researchers have hypothesised that these microorganisms can influence the phenotypic plasticity of animals, thereby enhancing their ability to thrive in diverse environments (Alberdi et al. 2016; Henry et al. 2021; Moeller and Sanders 2020; Rosenberg and Zilber-Rosenberg 2022; Voolstra and Ziegler 2020). For example, variability in gut microbiota is acknowledged as one of the most effective mechanisms for adapting to dietary changes (Teullet et al. 2023). Moreover, changes in gut microbiota have been linked to animal behaviour, as gut microorganisms are known to produce and metabolise neurotransmitters and hormones that can influence neurological processes (Cannas et al. 2021; Davidson et al. 2018; Noronha et al. 2024; Ezenwa et al. 2012; Forsythe et al. 2014; Johnson and Foster 2018; Mayer 2011; Strandwitz 2018). In fact, studies on domestic chickens and foxes kept in controlled conditions have demonstrated how certain microbiome features are associated with different behavioural responses to humans (Puetz et al. 2021, 2024), with higher taxonomic and functional diversities being associated with increased cognitive capacities and aggressive/fearful behaviour (Carlson et al. 2021; Sylvia et al. 2017; Agranyoni et al. 2021; Craddock et al. 2022). In addition, a recent study also showed that dogs suffering from generalised fear have different gut microbiomes and significant alterations of molecules

associated to GABA and glutamate neurotransmission, as well as bile acids metabolism, when compared to healthy dogs (Sacchettino et al. 2025). These observations raise the hypothesis that the differentiation in animal behaviour may be partially influenced by differences in the functional capacities of the gut microbial communities.

In this study, we analysed how the gut microbiota of cats changes in association with feralisation. Specifically, to explore their potential implications in facilitating the process and shaping behavioural changes in hosts, we used genome-resolved metagenomics to compare the taxonomic and functional features of gut microbiomes of 92 domestic and feral cats, sampled in six different locations across the world. With this data, we tested whether feral cats derived from domestic populations recovered an extended repertoire of microbial functions compared to domestic cats in the same region, and whether these changes were in line with the observed behavioural phenotypes.

2 | Materials and Methods

2.1 | Sample Collection

Veterinary clinics and shelters with trap-neuter-return programmes from six geographically separated countries (Aruba, Brazil, Cabo Verde, Denmark, Malaysia and Spain) participated in this study. In each location, faecal samples of cats defined as feral (i.e. individuals that had been originally captured in the streets/nature) and domesticated (i.e. individuals living with humans prior to entering the clinic or shelter) were collected (Table S1). To ensure standardised faecal sample collections, sampling kits were sent to each collaborator, including sampling instructions, cat metadata sheets, FTA cards (Whatman FTA Classic), sterile swabs, plastic bags and desiccant packs. FTA cards are specialised paper matrices used for the collection, storage and transport of biological samples, especially DNA, providing a convenient and stable solution for long-term storage without refrigeration. Upon contact with the sample, the FTA card lyses cells, denatures proteins and safeguards DNA from degradation, making it ideal for large-scale studies like this one, where standardised microbiota collection and preservation are essential (Bolt Botnen et al. 2023). The collaborator collected a fresh scat using a sterile swab to transfer some faecal material onto the FTA card, which was then left to dry. Two circles on the filter paper FTA card were collected for each cat faecal sample. The dry FTA card was then stored in a plastic bag until all samples were collected.

In addition to recording each cat's origin, information was collected on behavioural traits towards humans, age, sex, date of entry to the facility, date of sampling and use of antibiotics in the animal in each facility. Behavioural assessments were conducted immediately upon each cat's arrival at the clinic/shelter and at the initial point of contact with the assigned behavioural assessor. Key behavioural traits of interest included bites, hissing, avoidance and retreat from human presence. The assessor systematically documented each cat's response during initial contact to capture the unconditioned behavioural response to a novel human encounter. Once all data and samples were

collected, they were shipped to Denmark for storage at room temperature along with a new desiccant pack.

2.2 | DNA Extraction

DNA was extracted from 15 or 16 feral and domestic cats from each sampling site following Bolt Botnen et al. (2023). Briefly, approximately 1/4th of a circle was cut out by a scalpel and placed in a Monarch spin column. The 175 µL of elution buffer was added to each spin column before incubating at 37°C for 1 h. After incubation, each tube was centrifuged at 13,000×g for 10 min. The supernatant was placed in a new Eppendorf tube, ready for storage at −20°C or bead purification with SPRI beads. Bead purification followed the AMPure protocol for a DNA:Bead ratio of 1:1.4 and eluted in 50 µL of elution buffer. DNA was fragmented to an average length of 400 nucleotides using a Covaris M220 ultrasonicator with 50 µL DNA in the microTUBE-50 (Pulver power 30, duty factor 20, cycles per burst 50, treatment time 60 s).

2.3 | Library Building

DNA extracts were quantified using a Qubit and then diluted to approximately 100 ng in 16 µL for the library building. Libraries were built using the BEST ligation-based library preparation protocol (Carøe et al. 2018). Per reaction, the end repair master mix contained 2 µL T4 DNA ligase buffer, 1 µL reaction enhancer, 0.5 µL T4 PNK, 0.2 µL T4 polymerase and 0.2 µL dNTP 25 mM. The 3.9 µL of the master mix was added to each sample before incubating at 20°C for 30 min and 65°C for 30 min. The ligation step master mix contained 3 µL PEG 4000 50%, 0.5 µL T4 DNA ligase buffer and 0.5 µL T4 DNA ligase, per reaction. To each sample 1 µL adapter (50 µM) was added followed by 4 µL of the ligation master mix. The samples were then incubated at 20°C for 30 min and 65°C for 10 min. Fill-in master mix contained, per reaction, 3 µL ddH₂O, 1 µL Isothermal buffer, 0.2 µL dNTP 25 mM and 0.8 µL Bst 2.0 polymerase. The 5 µL of fill-in master mix was added to each sample followed by incubation at 65°C for 15 min and 80°C for 15 min. Each library was then diluted 1:2 with molecular-grade water. Bead purification followed the AMPure purification protocol, substituted with SPRI beads, at a ratio of 1:1.4 and eluted in 30 µL of elution buffer.

Dilutions (1:100) of each library were prepared for qPCR. A qPCR master mix was prepared with 11.8 µL ddH₂O, 2.5 µL 10× buffer, 2.5 µL MgCl₂ 25 mM, 0.5 µL BSA, 0.5 µL SYBR, 0.2 µL dNTP 25 mM, 0.5 forward primer (BGI-F 10 µM) and 0.5 µL TaqGold, per reaction. Each reaction consisted of 10 µL diluted DNA, 1 µL reverse primer (BGI-R 10 µM) and 19 µL master mix. qPCR settings were as follows: 95°C for 10 min; 30 cycles of 95°C for 20 s, 60°C for 30 s and 72°C for 40 s; followed by a final phase of 72°C for 7 min and a 4°C hold. The cycle threshold determined the number of cycles required for indexing. For indexing, a master mix was prepared with 12.3 µL ddH₂O, 2.5 µL 10× buffer, 2.5 µL MgCl₂ 25 mM, 0.5 µL BSA, 0.2 µL dNTP 25 mM and 0.5 µL TaqGold, per reaction. Each reaction consisted of 10 µL undiluted DNA, 1 µL of each indexed primer (BGI-R and BGI-F 10 µM) and 19 µL master

mix. PCR settings were as follows: 95°C for 10 min, 7–11 cycles of 95°C for 20 s, 60°C for 30 s, 72°C for 40 s, followed by a final 72°C for 7 min and a 4°C hold. Libraries were then bead purified with a ratio of 1:1.4 and eluted in 30 µL elution buffer and then quantified on the Qubit and 2–3 µL aliquoted for fragment analysis. Libraries were pooled together based on country. Due to practical reasons, libraries from five locations were sequenced on a DNBseq platform at BGI Denmark, while one location (Denmark) was sequenced using Illumina technology at a NovaSeq 6000 platform at Novogene (UK), after ensuring that both platforms are compatible for combined microbiome analyses (Mak et al. 2017; Smith et al. 2019).

2.4 | Bioinformatics

Using a custom snakemake-based (Mölder et al. 2021) pipeline, paired-end reads were trimmed and quality controlled using fastp v0.20.1 (Chen et al. 2018), with the following options: `—trim_poly_g`, `—trim_poly_x`, `—n_base_limit 5`, `—qualified_quality_phred 20`, `—length_required 35`. Processed reads were then mapped to the concatenated host reference genome assemblies (Human; GRCh38.p13), (Cat; F.catus_Fca126_mat1.0) using Bowtie2 (Langmead and Salzberg 2012) and samtools (Li et al. 2009) (default settings). Unaligned reads from each geographic location were then pooled and coassembled per location using metaSPAdes (Nurk et al. 2017) with the following kmer sizes: 21, 29, 39, 59, 79, 99, 119. The locality-wise coassembly approach was implemented for two key reasons: to minimise the risk of introducing artificial differences between the two cat origins, and to maximise sequencing depth, thereby enhancing coverage of the broadest possible microbial diversity. Coassembled contigs shorter than 1500 bp were removed. Each sample's reads were then mapped to its appropriate coassembly using Bowtie2. The resulting BAMs were used as input to MetaWRAP's (Uritskiy et al. 2018) binning module, and the coassembled contigs binned using CONCOCT (Alneberg et al. 2014), MaxBin2 (Wu et al. 2016) and MetaBAT2 (D. D. Kang et al. 2019). The output bins were automatically refined using MetaWRAP's bin_refinement module, with a minimum CheckM (Parks et al. 2015) completeness score of 70%, and a minimum checkM contamination score of 10%. dRep (Olm et al. 2017) was then used to dereplicate the refined bins, first into primary clusters > 90% average nucleotide identity (ANI) using MASH (Ondov et al. 2016), then into secondary clusters > 98% ANI with ANImf (Kurtz et al. 2004; Richter and Rosselló-Móra 2009). The non-host quality-controlled reads were then mapped against this dereplicated MAG catalogue as above. The resulting BAMs were then profiled using CoverM (<https://github.com/wwood/CoverM>) to create the final sample count table. The dereplicated MAGs were taxonomically annotated using the GTDB-tk (Chaumeil et al. 2019; Parks et al. 2020), which uses pplacer (Matsen et al. 2010), prodigal (Hyatt et al. 2010), HMMER3 (Eddy 2011), FastANI (Jain et al. 2018) and FastTree (M. N. Price et al. 2010). The MAGs were also functionally annotated using the DRAM pipeline (Shaffer et al. 2020), which searches predicted proteins against multiple databases (El-Gebali et al. 2019; Kanehisa et al. 2017; Rawlings et al. 2010; Suzek et al. 2015) using MMseqs2 (Steinegger and Söding 2017). Finally, gene annotations were distilled into Genome-Inferred Functional

Traits (GIFTs) using distillR (<https://github.com/anttonalberdi/distillR>), producing biologically meaningful annotations that highlight each bacterial genome's potential to degrade or synthesise compounds relevant to host metabolism. DistillR utilises a curated database of over 300 metabolic pathways, employing KEGG and Enzyme Commission (EC) identifiers to calculate standardised GIFT values. These values range from 0 to 1, where 0 signifies the absence of all genes associated with a particular metabolic pathway, and 1 indicates the presence of all necessary genes. For example, if a pathway step requires two specific identifiers, it is deemed complete when both are present, half-complete when only one is present, and empty if neither is present.

2.5 | Statistics

2.5.1 | Behavioural Phenotypes in Domestic and Feral Cats

We follow the earlier definitions of feral cats (Crowley et al. 2020; Gosling et al. 2013), defining a feral cat as an un-owned specimen capable of surviving with or without direct human intervention, and additionally showing fearful or defensive behaviour upon human contact. In each location, cats captured in the streets or nature were classified as feral, while those living in a residence with their owners were classified as domestic. To ensure the origin of the cats correlated with the behavioural phenotype, we analysed the differences in cats' fearfulness or defensive behaviour towards humans using chi-squared tests.

2.5.2 | Alpha and Beta Diversities

Alpha and beta diversity analyses were performed within the Hill numbers framework (Alberdi and Gilbert 2019), using different combinations of orders of diversity (q) and regularity in the Hilldiv2 R package (<https://github.com/anttonalberdi/Hilldiv2>). 'Richness' refers to the neutral Hill number of $q=0$, which is limited to the counts of MAGs. 'Neutral' refers to the neutral Hill number of $q=1$, in which the MAGs are weighted according to their relative representation. In the case of alpha diversity, it equals the exponential of the Shannon index, also known as Shannon diversity. 'Phylogenetic' refers to the phylogenetic Hill number of $q=1$, which accounts for the phylogenetic relationships between MAGs derived from the phylogenetic tree. 'Functional' refers to the functional Hill number of $q=1$, which accounts for the functional distances between MAGs derived from the GIFT table. Compositional differences were quantified using pairwise dissimilarity estimation via the Jaccard-type turnover metric (S) derived from the Hill numbers beta diversity. All four metrics were employed for alpha and beta diversity comparisons. Differences in alpha diversity between locations were assessed using linear models, while differences between origins were tested using linear mixed models with location as a random effect. PERMANOVA analysis was performed to compare the beta diversity differences between locations and origins using `vegan::adonis2` function, both considering the interaction between origin and location, as well as only considering the effect of origin, using location as strata, to ensure that

permutations happen within each sampling, preserving the site-level grouping.

2.5.3 | Hierarchical Modelling of Species Communities

To understand taxonomic and functional responses to cat origin and behaviour, we built two models of species communities using Hierarchical Modelling of Species Communities (HMSC), a multivariate hierarchical generalised linear model that uses Bayesian inference (Tikhonov et al. 2020). The main predictor of the first model was the categorical variable origin (domestic vs. feral), as defined by the provenance of the animals. The main predictor of the second model was the continuous variable behaviour, ranging from 0 (tame) to 1 (aggressive), calculated as the sum of four binary behavioural traits (bites, retreats, hisses and avoidance) that are commonly employed to assess aggressiveness in cats (Bennett et al. 2017; Frank and Dehasse 2003). Both models included the categorical covariate sex (male, female, unknown) and the log-transformed sequencing depth as a continuous covariate. In both cases, the response variable was the genome-size normalised and log-transformed counts of MAGs. Geographic location was included as a random effect to account for the non-independence of samples collected from the same location. Additionally, we included the MAG phylogeny to quantify the degree of phylogenetic signal in the MAG's responses to the variables.

We fitted the models assuming default priors and sampled the posterior distribution running four Markov Chain Monte Carlo (MCMC) chains, each running for 3750 iterations with 1250 discarded as burn-in. We thinned by 10 to obtain a total of 250 posterior samples per chain and 1000 total posterior samples. To test for MCMC convergence we measured the potential scale reduction factor for the beta (response to variables) and rho parameters (phylogenetic signal). We then performed five-fold cross-validation to evaluate the predictive performance of the models for the analysed MAGs in terms of R^2 . After fitting the first model, we quantified the posterior estimates for the binary variable 'origin' to identify MAGs that were significantly associated with either feral or domestic origins. In the case of the second model, we quantified the posterior estimates for the continuous variable 'behaviour' to identify MAGs that exhibited a significant positive or negative response along the behavioural gradient. The significance level was defined as having 90% of the posterior credible interval of the beta parameter, measuring the effect of either origin or behaviour, entirely positive or entirely negative (i.e. significance threshold of 0.9 posterior probability). To minimise the risk of false positives, we estimated functional differences between microbial communities associated with cats with different origins and behaviour, comparing the community-level functional attributes of a subset of MAGs with the highest predictive capacity and explanatory power. Specifically, for each MAG, we multiplied the R^2 between the observed and predicted values revealed by the cross-validation with the percentage of variation explained by the variable of interest as yielded by the variance partitioning analysis. We then selected a threshold value of 0.005 that maximises the number of MAGs significantly associated with the variables of interest, while excluding the MAGs that do not exhibit changes in

response to origin or behaviour. This approach minimised the risk of bias in the functional analyses from MAGs that our models could not accurately predict.

3 | Results

3.1 | A Global Catalogue of Cat-Associated Microbial Genomes

We generated 298.1 GB of shotgun sequencing data (3.3 ± 1.4 GB/sample) from faecal samples of 96 cats collected across six countries (Figure 1a; Table S2). Four samples were removed from the analysis due to low library and data quality. Metagenomic assembly and binning of this dataset yielded a total of 229 non-redundant metagenome-assembled genomes (MAGs) (Figure 1b; Table S3). The average CheckM completeness of these MAGs

was $90.02\% \pm 7.79\%$, with an average CheckM contamination of $1.32\% \pm 1.18\%$ (Figure S1).

The MAGs were taxonomically assigned to 10 bacterial phyla (Figure 1b, Table S4). Despite their phylogenetic relatedness, MAGs belonging to Bacillota A displayed extensive functional variability, covering over half of the functional space in the functional ordination (Figure S2). Bacterial communities were dominated by Bacteroidota ($29.82\% \pm 17.98\%$), Bacillota A ($20.34\% \pm 13.17\%$) and Actinomycetota ($18.39\% \pm 16.05\%$), with the remaining bacterial phyla with average value with less than 9% (Figure 1c). At the family level, Bacteroidaceae ($27.09\% \pm 17.68\%$), Lachnospiraceae ($12.12\% \pm 10.54\%$), Coriobacteriaceae ($9.56\% \pm 9.52\%$), Helicobacteraceae ($6.39\% \pm 12.29\%$) and Bifidobacteriaceae ($4.37\% \pm 7.20\%$) were the five most abundant in the gut microbiome of cats (Figure S3). Furthermore, *Prevotella* ($16.47\% \pm 15.19\%$), *Collinsella*

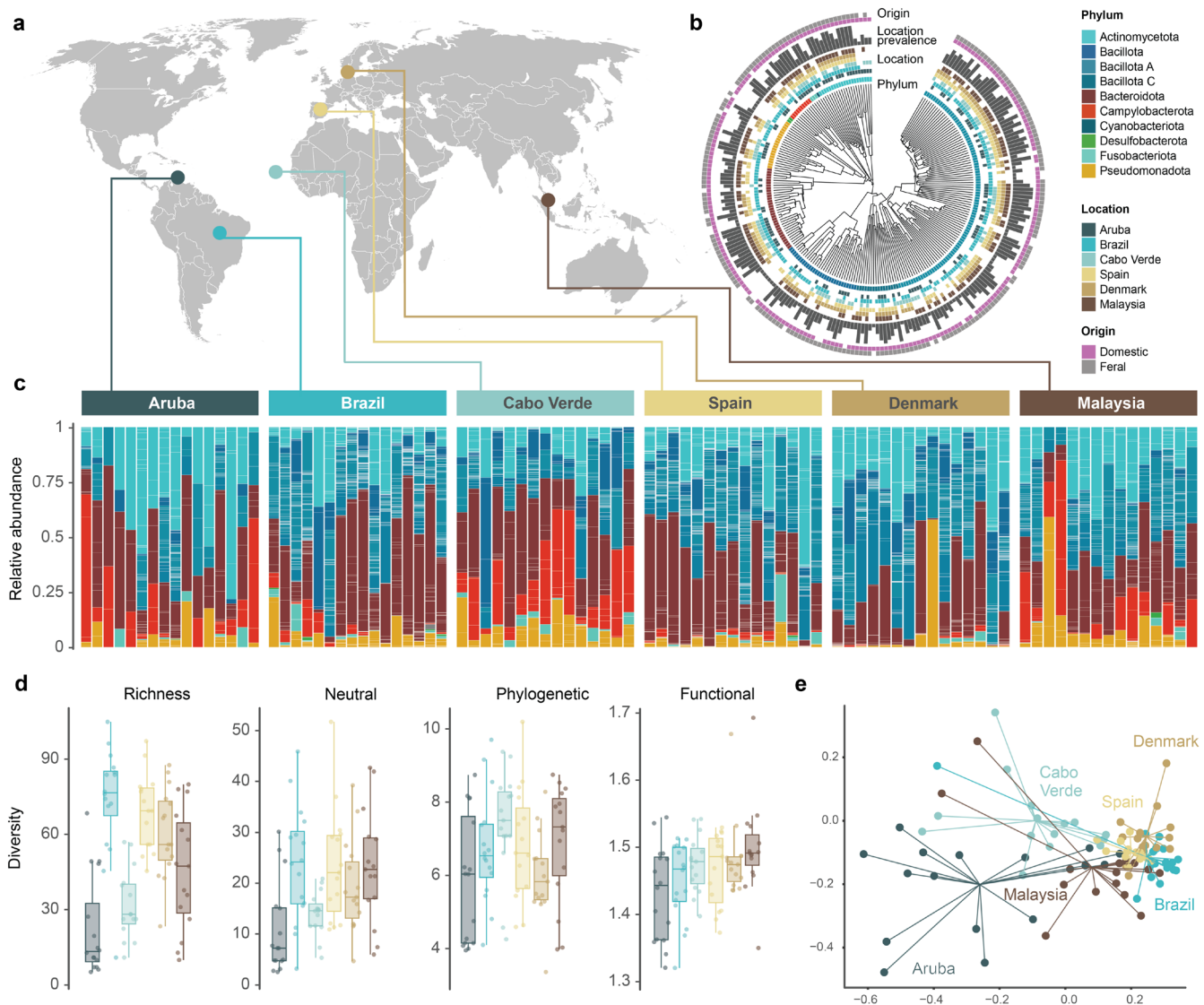


FIGURE 1 | Overview of the geographic locations and genome catalogue characteristics. (a) World map with the geographic locations of the samples. (b) Reference metagenome-assembled genome (MAG) catalogue generated from the cat faecal samples, with outer rings indicating detection of MAGs in different geographic regions and origin (domestic or feral). (c) Taxonomic composition of the faecal samples, each tile representing a MAG coloured by taxonomic phyla. (d) Alpha diversity differences across geographic regions based on four different Hill numbers diversity metrics. (e) Ordination of richness-based compositional differences across samples, coloured and grouped by location.

(9.56% $\pm$ 9.52%), *Phoecaicola* (5.36% $\pm$ 6.66%), *Helicobacter B* (5.21% $\pm$ 11.59%) and *Bifidobacterium* (4.37% $\pm$ 7.2%) the five most abundant bacterial genera (Figure S4).

3.2 | Differences Across Localities

While 66 MAGs (29%) were present across all locations, the rest were specific to one or a few locations (Figure S5). Brazilian samples exhibited the highest number of MAGs, including 20 unique MAGs, with Cabo Verde showing both the fewest total and unique number of MAGs. The geographic spread of bacteria was not affected by the lifestyle of the cats (MANN-WHITNEY; $W=25,706$, p -value=0.713), with MAGs from domestic and feral cats being present in 3.12 ± 1.93 and 3.19 ± 2.03 locations, respectively. Richness and neutral diversity metrics also varied across geographic locations (LM: $p < 0.001$ for both metrics), with Aruba and Cabo Verde exhibiting the lowest neutral diversities, yet similar phylogenetic and functional metrics as the rest of the locations (Figure 1d) (LM_{phylogenetic}: $p=0.06$; LM_{functional}: $p=0.05$, all pairwise comparison, $p > 0.05$).

The location of the cats explained most of the compositional variability of the gut microbiota (Figure 1e). Over 20% of the variability of the neutral (PERMANOVA; $R^2=0.221$, p -value=0.001) and phylogenetic (PERMANOVA; $R^2=0.268$, p -value=0.001) diversity was explained by the location and 58% of the functional variability (PERMANOVA; $R^2=0.584$, p -value=0.001). Spain and Denmark exhibit more similar microbial communities than the rest of the localities (pairwise-Permanova: $R^2=0.0972$, p -adjusted=0.075). Despite geographical differences, cats from Brazil had more similar gut microbiota to cats from Spain (pairwise-Permanova: $R^2=0.0630$, p -adjusted=0.555) than from the nearby Aruba (pairwise-Permanova: $R^2=0.1377$, p -adjusted=0.015). Cats from Aruba, Cabo Verde and Malaysia exhibited a more distinct microbial community than those from Denmark, Brazil and Spain, which clustered closely together (Figure S6).

3.3 | Taxonomic and Functional Microbiome Differences Between Domestic and Feral Cats

We then compared whether microbiome features differed according to the feral or domestic origin of the cats. We observed that overall alpha (Mixed models, $p > 0.05$) and beta diversity (PERMANOVA, $p > 0.05$) analyses, when controlled for location, did not yield any significant differences in microbiome diversity and composition between feral and domestic cats (Figure 2a,b). Accordingly, the only bacteria unique to either of the two origins were rare species found in just one location, with no widespread MAGs specific to either domestic or feral origins. However, when considering relative abundances, hierarchical modelling of species communities (HMSC) revealed that 61 MAGs were associated with domestic cats while 34 MAGs were associated with feral cats (Figure 3a, Figure S7). MAGs belonging to a range of bacterial phyla were differentially abundant in domestic cats, including Pseudomonadota (11 MAGs from CAG-239, Burkholderiaceae and Succinivibrionaceae), Campylobacterota (3 MAGs from Helicobacteraceae), Bacteroidota (9 MAGs from Bacteroidaceae), Actinomycetota (18 MAGs from

Actinomycetaceae, Atopobiaceae, Bifidobacteriaceae, Coriobacteriaceae, Mycobacteriaceae), Bacillota A (11 MAGs from Butyricicoccaceae, Oscillospiraceae, Peptoniphilaceae and Ruminococcaceae) and Bacillota C (8 MAGs from Acidaminococcaceae, Dialisteraceae, Megasphaeraceae), Desulfobacterota (1 MAG from Desulfovibrionaceae). In feral cats, MAGs belong to Bacillota A (14 MAGs from Acutalibacteraceae, Anaerotrignaceae, Clostridiaceae, Lachnospiraceae, Peptostreptococcaceae and UBA1381), Bacillota (17 MAGs from Enterococcaceae, Lactobacillaceae and Streptococcaceae) and Fusobacteriota (3 MAGs from Fusobacteriaceae).

Using our HMSC model, we predicted the typical microbial community compositions for feral and domestic cats, while controlling for sex and location. This enabled us to estimate the community-level functional differences of the microbiota associated with both conditions (Figure 2c,d). The microbial community of domestic cats exhibited a higher capacity to degrade polysaccharides such as mixed-linkage glucans (Figure 3c). The serine and tyrosine amino acid biosynthetic capacity was also higher in domestic cats, and the capacity to biosynthesise the aromatic compound indole-3-acetate, a major plant growth hormone. The overall repertoire of enriched functions was considerably wider in the case of feral cats. These included an enhanced capacity for biosynthesising the amino acids glutamate, cysteine and proline and the amino acid derivative spermidine. The capacities to produce the SCFA acetate, propionate and butyrate and vitamin E (Tocopherol/tocotorienol) and B12 were also increased. The catabolic repertoire of enriched functions listed capacities for degrading the amino acids histidine, tryptophan, proline and threonine, the lipids triglyceride, oleate, fatty acid and dicarboxylic acids, the antibiotics streptogramin, lincosamide and macrolide, the sugar sucrose, D-mannose and lactose, various nitrogen compounds such as methylamine and allantoin, the alcohol 2,3-Butanediol and two xenobiotics, such as toluene and xylene.

3.4 | Taxonomic and Functional Microbiome Differences Across Behavioural Phenotypes

The behavioural phenotypes observed during animal sampling demonstrated a clear correlation with the cats' domestic or feral origins (Table S5). Specifically, feral cats showed significantly higher levels of avoidance ($X^2=15.5$, $df=1$, p -value > 0.001), hissing ($X^2=11.2$, $df=1$, p -value < 0.001), biting ($X^2=11.8$, $df=1$, p -value < 0.001) and retreating ($X^2=5.44$, $df=1$, p -value=0.0197) behaviours towards humans compared to domestic cats (Figure S8). To further explore the relationship between the microbiome and behaviour, we conducted a second HMSC analysis, using a quantitative index that ranged from tame to aggressive behaviour as the primary variable. The model identified 25 microbial taxa associated with tame behaviour, of which 15 were also linked to domestic origin, whereas none were associated with feral origin (Figure 3b, Figure S9). This group included 8 of the 11 Pseudomonadota taxa linked to domestic origin. Conversely, another 25 taxa were associated with aggressive behaviour, though their connection to feral origin was less clear. While 10 taxa from Fusobacteria and Bacillota were linked to both feral origin and aggressive behaviour, five

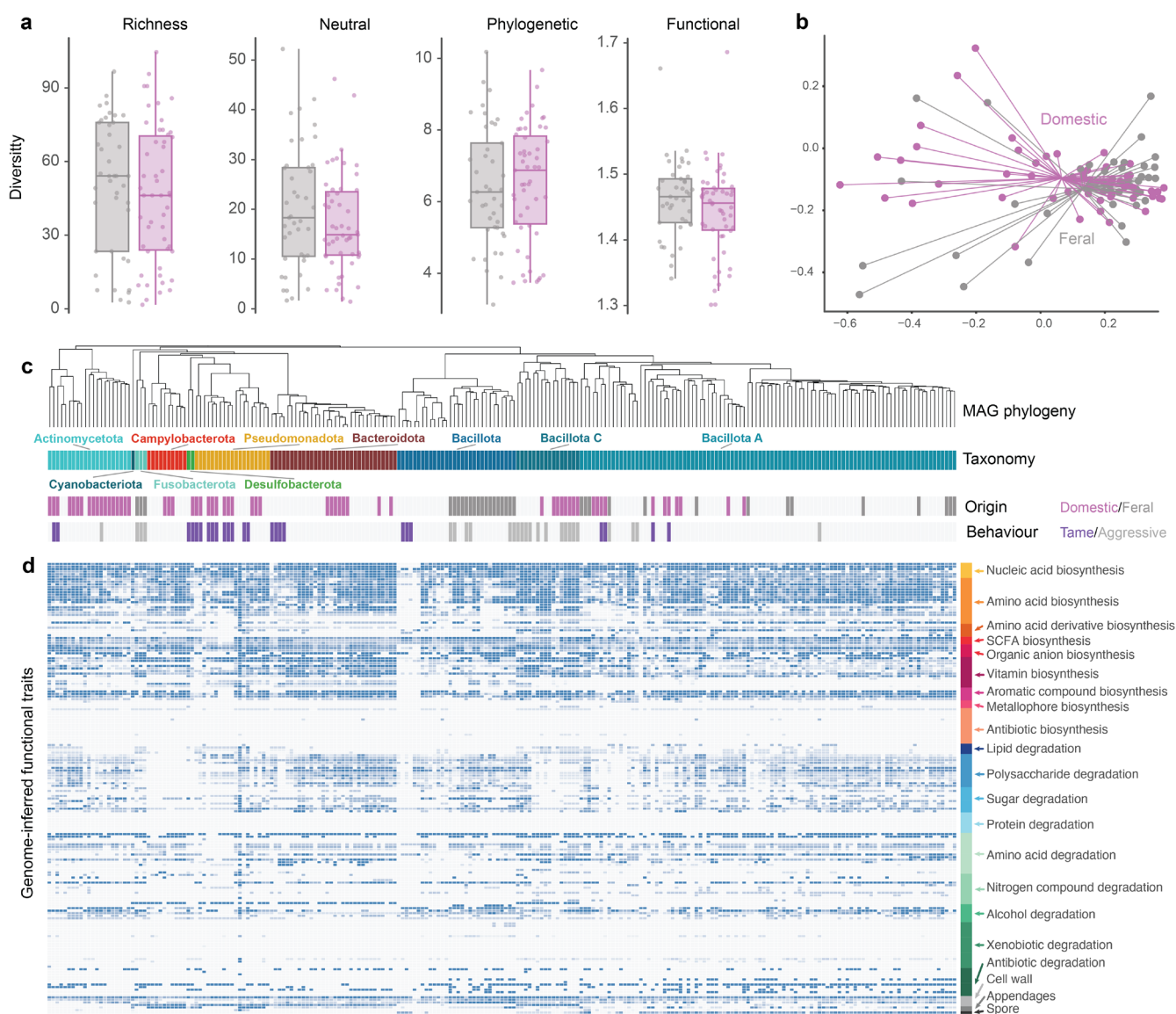


FIGURE 2 | Microbiota differences between domestic and feral cats. (a) Alpha diversity differences between domestic and feral cats, based on four different Hill numbers diversity metrics. (b) Compositional differences between domestic and feral cats, samples coloured by domestic or feral origin. (c) Phylogenetic tree of reconstructed genomes, highlighting genomes significantly associated with specific origins and behaviours. (d) Heatmap of 170 genome-inferred functional traits, with blue colour indicating higher capacity to perform a given metabolic function. Metagenome-assembled genomes (MAGs) are phylogenetically sorted, with their respective enrichment levels annotated above the heatmap. GIFTs are sorted according to their functional group.

Megasphaeraceae bacteria linked to aggressive behaviour were also associated with domestic origin. Comparative analysis of the Megasphaeraceae abundances between the few aggressive domestic animals against tame counterparts did not yield significant differences (ANCOMBC: p -values > 0.05). However, we identified two of the most aggressive individuals with exceptionally high levels of *Megasphaera elsdenii*, which may have skewed the signal.

Functional predictions from the behaviour modelling indicated that a greater number of metabolic functions were positively associated with aggressive behaviour compared to tame behaviour, mirroring the observed differences between feral and domestic origins (Figure 3d). Many of the functions associated with feral origin were also identified as associated with

aggressive behaviour. These included the biosynthesis of multiple SCFAs and the amino acid derivative spermidine among others. Additionally, production capacities for lactate, propionate and acetate, as well as degradation of threonine, serine, arginine and leucine, increased with the aggressive phenotype. Production of vitamins B12, B5 and B7 were also positively associated with aggressive behaviour. In contrast, we observed no concordance between microbial functions associated with tameness and domestic origin, as none of the 11 functions linked with tame behaviour were also related to domestic origin.

To further explore the relationship between taxa and functions, we performed a principal coordinates analysis (PCoA) to ordinate bacterial genomes based on their functional features. This was followed by a visualisation of bacteria associated with

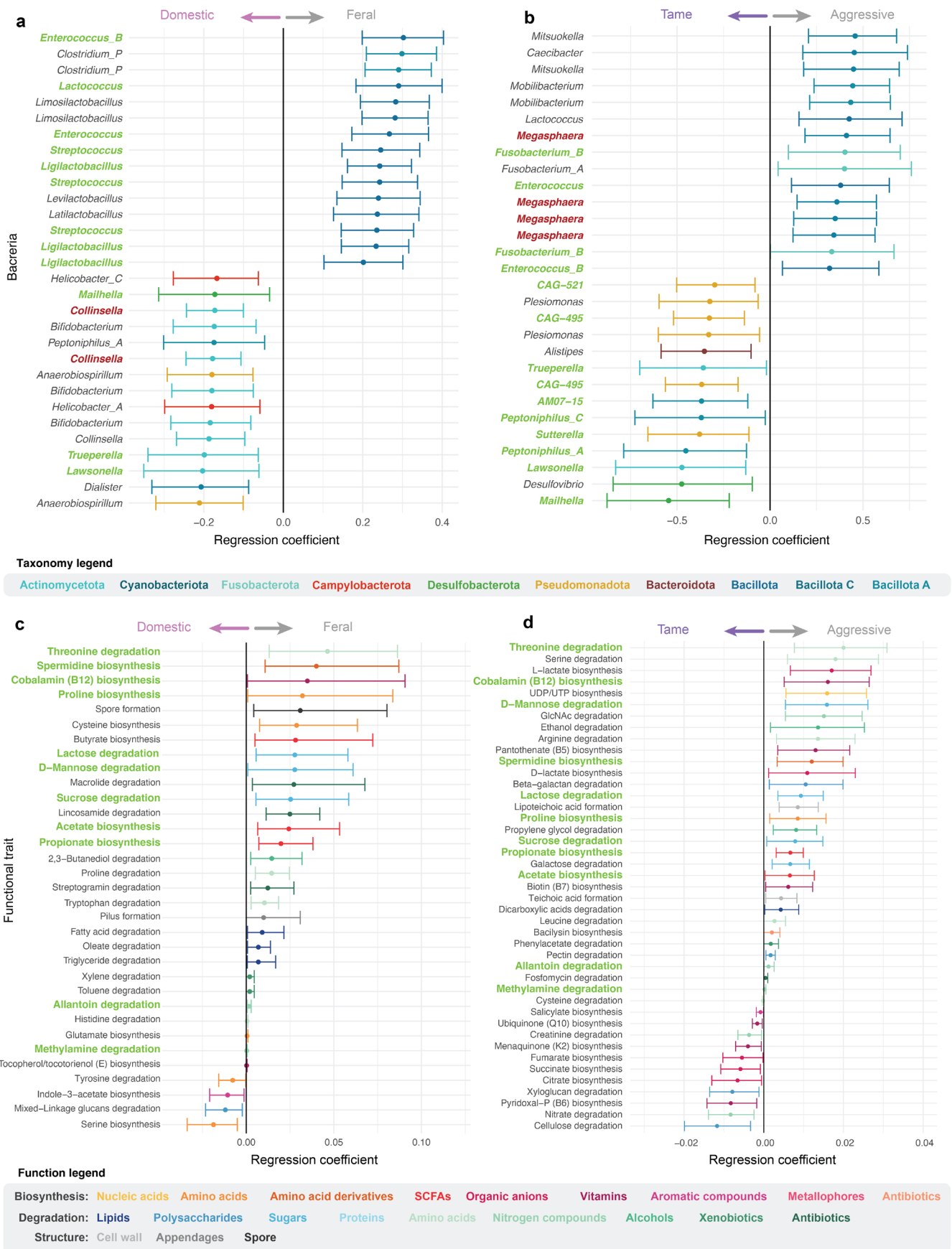


FIGURE 3 | Legend on next page.

FIGURE 3 | Taxonomic and functional differences according to cat origin and behaviour. (a) Top-15 genomes associated with domestic and feral origins. (b) Top-15 genomes associated with tame and aggressive behaviours. (c) Functional traits associated with domestic and feral origins. (d) Functional traits associated with tame and aggressive behaviours. The taxa and functions coloured in green indicate congruence between origin and behaviour (i.e. domestic-tame, feral-aggressive), while the coloured in red indicate incongruence between origin and behaviour (i.e. domestic-aggressive, feral-tame).

either aggressive or tame behaviour (Figure 4a). The analysis revealed that bacteria—predominantly from Bacillota—linked to aggressive behaviour exhibited significantly greater metabolic capacities for functions identified as associated with aggressive behaviour, compared to those associated with tame behaviour (Figure 4b).

4 | Discussion

Our global genome-resolved metagenomic data set provides unique insights into the taxonomic and functional diversity of cat-associated microbiomes across different geographies and lifestyles. We reconstructed 229 bacterial genomes, with 42 (18%) displaying ANI values below 95% compared to the closest representatives in the GTDB database, indicating potential new species (Rodriguez-R et al. 2024). The gut microbiome of cats was predominantly composed of Bacteroidota, Bacillota A and Actinomycetota, with notable presences of Campylobacterota and Pseudomonadota, reflecting a typical microbiome profile of a carnivorous mammal (Milani et al. 2020; Zoelzer et al. 2021). Nearly one-third of the bacteria detected in the faecal samples were present in all six sampled locations, indicating a globally distributed core microbiota spanning eight phyla. The remaining bacteria were found in only one or a few locations, resulting in substantial compositional variation across different geographies, consistent with previous observations in other domestic animals, such as horses (Ang et al. 2022) and dogs (Yarlagadda et al. 2022). We acknowledge that the sequencing strategy, which for practical reasons pooled all libraries from a given country, may have accentuated differences across geographical locations. However, we argue that this approach helped minimise design-based biases, as geographical location was treated as a random effect in modelling microbiome associations with origin and behaviour—the two primary objectives of the study. Furthermore, consistent with previous findings (Mak et al. 2017; Smith et al. 2019), sequencing Danish samples using Illumina technology instead of DNBseq did not introduce significant biases.

4.1 | Domestic and Feral Cats Exhibit a Taxonomic Footprint of Lifestyle Origin

Despite the significant differences observed across geographic regions, hierarchical modelling of microbial communities revealed taxonomic differences between domestic and feral cats once these regional variations were accounted for. The number of differentially abundant bacterial taxa in domestic and feral cats was similar, corresponding to the fairly balanced alpha diversities in microbiomes from both origins. However, the taxonomic profiles of these taxa differed. Bacteria belonging

to Pseudomonadota, Bacteroidota, Campylobacterota and Desulfobacteriota were more abundant in domestic cats. The bacteria with the strongest associations with domestic origin belong to seven phyla. In contrast, the microbiome of feral cats was characterised by a higher abundance of many Bacillota and Fusobacteriota species. While some Bacillota species found linked to feral origin, such as *Limosilactobacillus reuteri*, *Latilactobacillus sakei*, *Ligilactobacillus agilis* and *L. animalis*, are often employed as probiotics due to their capacity to produce lactic acid (Abuqwider et al. 2022; Boll et al. 2024; Nawaz Khan et al. 2024; Yu et al. 2024), many Fusobacteria are pathogenic to humans (Roberts 2000; Robinson et al. 2020). *Megasphaera* and *Prevotella* were differentially abundant in domestic cats, along with increased *Clostridium* and *Fusobacterium* in feral cats. These observations align largely with previous studies evaluating the impact of dietary transitions from a kibble diet high in carbohydrates to a more high-protein raw diet in cats and dogs (Bermingham et al. 2017; Butowski et al. 2019; Hooda et al. 2013; Lubbs et al. 2009). Moreover, *Bifidobacterium* and *Dialister*, which were differentially abundant in domestic cats in our study, were identified as the most featured genera in obese cats distinguishing them from normal cat microbiota (Ma et al. 2022). In consequence, these shifts in the microbial community likely reflect the differences in diet and human exposure between domestic and feral cats.

4.2 | Feral Cat Microbiomes Have Increased Functional Capacities Compared to Domestic Cats

Beyond taxonomic differences, our bioinformatic and statistical analyses enabled us to capture differences in the functional repertoires of bacteria associated with domestic and feral cats, shedding light on the pathways potentially affecting the domestication and feralisation processes. Of the 170 genome-inferred functional traits (GIFTs) analysed, only four showed consistent patterns of enrichment in domestic cats. The reduced number of enriched functional capacities in domestic animals may be associated with their consumption of commercial cat food, which is highly digestible and supplemented with rapidly absorbed nutrients (Deng et al. 2019). This likely leads to less substrate available for bacterial activity, depleting the functional repertoire of microbial communities and potentially diminishing their contribution to host biology.

In contrast, we found higher values of 30 GIFTs in feral cats. These functional traits encompassed many biosynthetic and degradation pathways, suggesting a more significant role of gut microbiota in feral cats compared to domestic ones. These findings are consistent with earlier observations that domestic animals tend to harbour simplified microbial communities, exhibiting reduced functional potential compared to their wild counterparts (Alessandri et al. 2019). A meta-analysis of

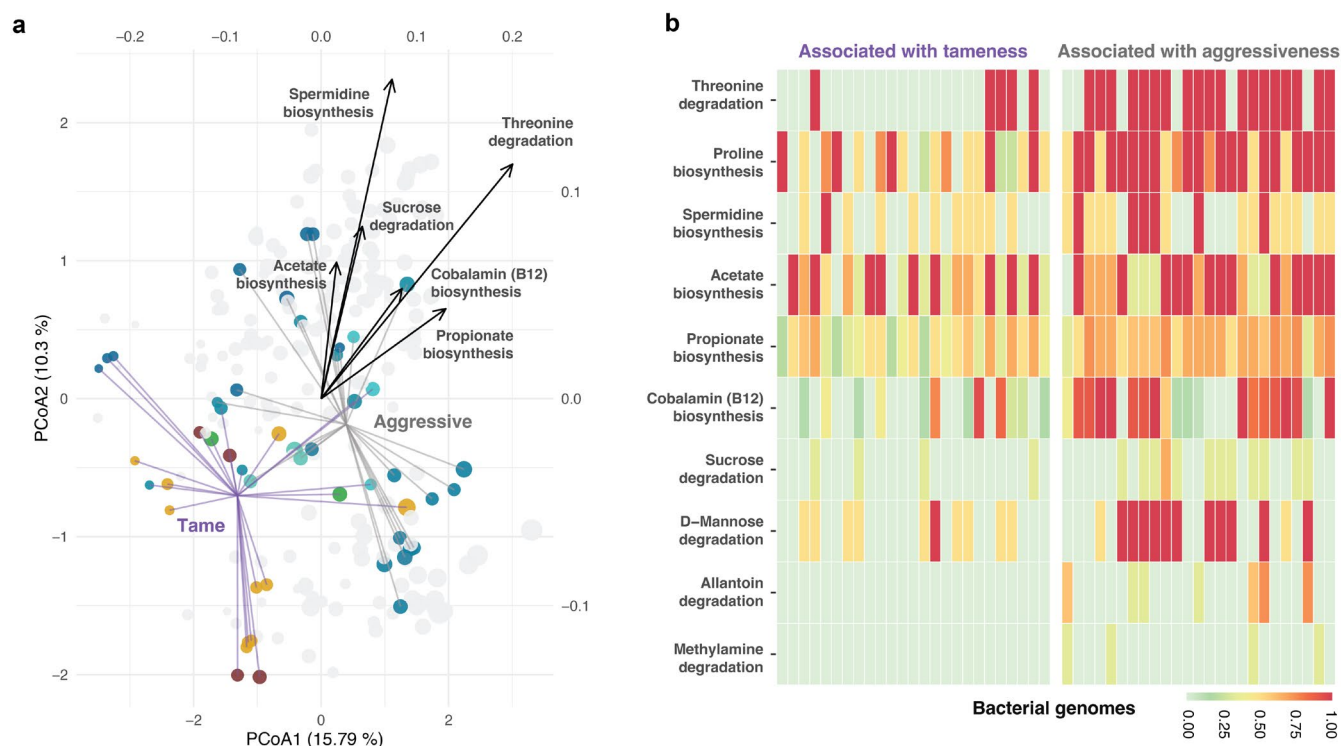


FIGURE 4 | Functional properties of bacteria associated with cat behaviour. (a) Principal Coordinate Analysis of the bacterial genomes based on their functional characteristics. The bacteria associated with tameness are clustered together with purple lines, while bacteria associated with aggressiveness are clustered with grey lines. Bacteria with no significant association with behaviour are displayed without colour, while the bacteria associated with behaviour are coloured according to phylum (see Figures 2 and 3 for legend). The eigenvectors indicate the directionality and intensity of the six most relevant functions. (b) Functional traits of genomes associated with tameness and aggressiveness.

the dietary habits of feral cats confirmed that they are obligate carnivores (Plantinga et al. 2011), consuming a diverse range of prey, including rodents, birds, fish and insects, along with a significant intake of human garbage (Doherty et al. 2015), which may also explain the higher capacities of antibiotic degradation, including those for macrolides, lincosamides and streptogramins. It has been estimated that 98% of the daily energy intake derives from crude protein and fat (Plantinga et al. 2011). Accordingly, we observed higher capacities of many microbial pathways associated with the degradation of such dietary components. Specifically, the microbiome of feral cats showed an enhanced capacity to break down several amino acids and derivatives, as well as lipids, including dicarboxylic acids, fatty acids, oleate and triglycerides.

4.3 | Higher Microbial Functional Capacities Align With Cat Behaviour

The taxonomic and functional differences observed between microbial communities of domestic and feral cats not only reflect dietary changes but also align with the behavioural distinctions seen between these two groups, particularly the fearful and defensive behaviour feral cats exhibited when in contact with humans, in contrast to the docile behaviour observed in domestic cats. This is consistent with studies showing that microbiota can indeed influence animal behaviour (Kamimura et al. 2024; Sudo et al. 2004; Warda et al. 2019; Watanabe et al. 2021; X. Zhang et al. 2022). As feralisation and domestication are rapid

processes involving drastic changes in an animal's environment and diet, they may trigger microbiota shifts that could alter the microbiota-gut-brain interaction.

Our second modelling using behavioural phenotypes as primary predictors highlighted that the taxonomic congruence of bacteria associated with feral origin and aggressive behaviour was much more marked than the congruence observed between domestic and tameness, suggesting that aggressiveness is more closely linked to a feral lifestyle than tameness is to a domestic one. The only two genera exhibiting contrasting patterns were *Collinsella*, associated with domestic but aggressive cats, and *Megasphaera*, associated with feral but tame cats. Both types of bacteria are among the most abundant bacteria in the core microbiome of cats (Ganz et al. 2022). In addition, various members of the order Veillonellales and Lactobacillales were associated with aggressive behaviour, mirroring previous observations related to increased fear and anxiety in dogs, chickens and horses (Bulmer et al. 2019; Mondo et al. 2020; Puetz et al. 2021; X.-Y. Zhang et al. 2018; Zhou et al. 2020). While these behavioural traits are generally selected against in domestic animals, they may provide an anti-predatory advantage for feral cats.

While most of the referenced publications relied on taxonomic profiling derived from 16S rRNA amplicon sequencing, a major advantage of the genome-resolved metagenomics approach we employed lies in its ability to provide direct functional insights from bacterial genomes reconstructed from metagenomic data. Unlike taxonomic approaches that can be impacted by

strain-specific functional variability, genome-resolved metagenomics enables direct reconstruction and analysis of functional pathways from bacterial genomes, providing a more accurate assessment of metabolic potential that may influence behaviour (Koziol et al. 2023).

In line with the functional microbiome signatures of cat origin, microbiome modelling according to animal behaviour yielded more metabolic functions positively associated with aggressiveness than tame behaviour. Functional analyses did not yield a strong congruence between tame behaviour and domestic origin, with the only related functions being the capacities for degrading cellulose and xyloglucan observed among tame cats and the capacity to degrade mixed-linkage glucans associated with domestic origin. In contrast, the congruence between aggressive behaviour and feral origin was much stronger, sharing a total of 12 GFTs. Aggressive cats exhibited an enhanced ability to degrade simple sugars (e.g. mannose, lactose, sucrose and galactose), which were also associated with the feral origin. These variations in polysaccharide and sugar degradation pathways have significant implications for short-chain fatty acid (SCFA) production, which were also congruent. Among the biosynthetic pathways affected, capacities for producing acetate and propionate were notably higher in both aggressive and feral cats. Beyond their significance for animal health (Frost et al. 2014; Goswami et al. 2018; Hu et al. 2022; Rowe et al. 2024), SCFAs could also influence observed behaviours by altering brain function via immune, endocrine and vagal pathways, directly by crossing the blood–brain barrier or acting as endogenous ligands (Dalile et al. 2019; Kratsman et al. 2016; Mirzaei et al. 2021; Silva et al. 2020). As an example of their relevance in brain function, acetate has been shown to offer protection against cognitive impairment in mice (Erny et al. 2021; Zheng et al. 2021). Acetate can also be incorporated into the glutamate–glutamine transcellular cycle and influence hypothalamic neurotransmission, as well as acting as an epigenetic modulator that could have contributed to gene expression changes during domestication (Anastasiadi et al. 2022; Béltéky et al. 2018; Nätt et al. 2012). In fact, acetate synthesis capacity has been recently found to be higher in the microbiomes of foxes genetically selected for aggressive behaviour (Puetz et al. 2024), reinforcing the possible link between microbiome functions and behaviour in feral cats.

Mirroring the patterns of association with feral origin, our second model also yielded positive associations of aggressiveness with the capacity to degrade many amino acids, as well as to biosynthesise spermidine and cobalamin (vitamin B12). Commonly supplemented in commercial diets, vitamin B12 can modulate behaviour, its deficiency leading to cognitive impairment and a range of neurological disorders (W. K. Kang et al. 2024). Additionally, the enhanced capacities for synthesising amino acids and derivatives also hold the potential to influence brain function. The capacity to biosynthesise proline, an amino acid with a significant impact on psychotic disorders (Mayneris-Perxachs et al. 2022; Savio et al. 2012; Yao and Han 2022), was significantly higher in feral and aggressive cats. Similarly, the amino acid derivative spermidine has been recently shown to improve cognitive function (Flory et al. 2023; Schroeder et al. 2021). Higher intestinal bacterial degradation of threonine can partially reduce their availability in the brain

(Tovar et al. 1988), thereby influencing the production of key neurotransmitters such as glycine, which have important brain functions (Boehm et al. 1998).

4.4 | Insights Into the Role of Microbiomes in Feralisation and Domestication Processes

Feralisation is often recognised as the inverse process of domestication (E. O. Price 1984), where many changes induced by domestication are expected to revert (Henriksen et al. 2018). Despite the broad geographical range screened, our results consistently showed that feral cats gained more microbial functions than their domestic counterparts, confirming our prediction that feral cats would exhibit a higher functional repertoire compared to domestic cats. Community-level multivariate modelling unveiled significant functional changes in the absence of broad-resolution microbiota markers such as diversity metrics, highlighting the value of statistical tools such as HMSC for functional microbiome research (Koziol et al. 2023). These findings emerged despite the inability to determine whether feral cats were born in the wild or became feral during their lifetime, a factor that likely introduced noise and uncertainty. Greater control over the long-term origin of the animals would likely have enhanced the detected signals. Targeted gut microbiome manipulation (e.g. faecal microbiota transplants) combined with controlled behavioural experiments are necessary to establish causal relationships between microbiota, behaviour and feralisation (Martin Bideguren et al. 2024; Moeller and Sanders 2020). Nonetheless, our findings suggest that a shift in the microbiome might favour the acquisition of physiological and behavioural traits beneficial for a feral lifestyle, as well as a likely effect on the increased aggressiveness and elusiveness observed in the sampled feral cats, which could guide future experimental studies that test the causality between microbiome functioning and cat behaviour.

Author Contributions

O.A. and A.B.B. performed the research, analysed the data and wrote the paper. R.E. and I.O. analysed the data. L.S.B. and M.B.B. contributed to the data generation. M.T.P.G. designed the research. A.A. designed the research, analysed the data and wrote the paper. All authors reviewed and validated the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Sequencing data can be accessed through the BioProject PRJEB79769, with original raw files published under accession numbers ERR13636309-ERR13636404 (Table S2) and metagenome assemblies published under accession numbers ERZ25037487-ERZ25037492 (Table S3). Metadata tables, analysis code, ENA checklists and Data S1 and tests can be found in the dedicated Github repository as a Rmarkdown webbook (https://alberdilab.github.io/domestic_feral_cat_metagenomics), which was frozen in Zenodo with doi 10.5281/zenodo.13802251 (Alberdi 2024). Benefit Sharing: Benefits from this research accrue from the sharing of our data and results on public databases as described above.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.